BRAC University

Department of Mathematics and Natural Sciences

Lecture on

Real Analysis II (MAT321)

Metric in $\mathbb{R}^n$, $\mathbb{C}^n$ and Function Spaces

Metric, pseudo-metric and Examples

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Conducted by

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Motivation of Metric

Metric Space

Metric Space

A **metric space** is a pair (X, d) where $X \neq \varnothing$ and $d : X \times X \to \mathbb{R}$ satisfies, for all $x, y, z \in X$:

(M1) $d(x, y) \geq 0$;

(M2) $d(x, y) = 0 \iff x = y$;

(M3) $d(x, y) = d(y, x)$;

(M4) $d(x, z) \leq d(x, y) + d(y, z)$.

Here d is called a **metric** on X.

Pseudo-Metric Space

A **pseudo-metric space** is a pair (X, d) where $X \neq \varnothing$ and $d : X \times X \to \mathbb{R}$ satisfies, for all $x, y, z \in X$:

(PM1) $d(x, y) \geq 0$;

(PM2) $d(x, y) = d(y, x)$;

(PM3) $d(x, z) \leq d(x, y) + d(y, z)$.

Here d is called a **pseudo-metric** on X.

Metric on Real Line

Euclidean Metric on Real Line

Show that $d : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ defined as

$$d(x, y) = |x - y|$$

is a metric on $\mathbb{R}$.

Another Metric on Real Line

Show that $d : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ defined as

$$d(x, y) = 3|x - y|$$

is a metric on $\mathbb{R}$.

Problem

Show that $d : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ defined as

$$d(x, y) = |x - 2y|$$

is a not metric on $\mathbb{R}$.

Problem

Show that $d : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ defined as

$$d(x, y) = (x - y)^2$$

is a not metric on $\mathbb{R}$.

Theorem

Show that $d : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ defined as

$$d(x, y) = \sqrt{|x - y|}$$

is a metric on $\mathbb{R}$.

Metric on $\mathbb{R}^2$

Standard Euclidean Metric on $\mathbb{R}^2$

The metric space $\mathbb{R}^2$, called the *Euclidean plane*, is obtained by taking the set of ordered pairs of real numbers,

$$x = (\xi_1, \xi_2), \qquad y = (\eta_1, \eta_2),$$

and defining the *Euclidean metric* by

$$d(x, y) = \sqrt{(\xi_1 - \eta_1)^2 + (\xi_2 - \eta_2)^2}$$

Show that d is a metric on $\mathbb{R}^2$.

Taxicab Metric on $\mathbb{R}^2$

The metric space $\mathbb{R}^2$, called the *Euclidean plane*, is obtained by taking the set of ordered pairs of real numbers,

$$x = (\xi_1, \xi_2), \qquad y = (\eta_1, \eta_2),$$

and defining the *Euclidean metric* by

$$d(x, y) = |\xi_1 - \eta_1| + |\xi_2 - \eta_2|$$

Show that d is a metric on $\mathbb{R}^2$.

d_p Metric on $\mathbb{R}^2$

The metric space $\mathbb{R}^2$, called the *Euclidean plane*, is obtained by taking the set of ordered pairs of real numbers,

$$x = (\xi_1, \xi_2), \qquad y = (\eta_1, \eta_2),$$

and defining the *Euclidean metric* by

$$d_p(x, y) = (|\xi_1 - \eta_1|^p + |\xi_2 - \eta_2|^p)^{\frac{1}{p}} \qquad (p \geq 1).$$

Show that d_p is a metric on $\mathbb{R}^2$.

d_∞ Metric on $\mathbb{R}^2$

The metric space $\mathbb{R}^2$, called the *Euclidean plane*, is obtained by taking the set of ordered pairs of real numbers,

$$x = (\xi_1, \xi_2), \qquad y = (\eta_1, \eta_2),$$

and defining the *Euclidean metric* by

$$d_\infty(x, y) = \max\{|\xi_1 - \eta_1|, |\xi_2 - \eta_2|\}.$$

Show that d_∞ is a metric on $\mathbb{R}^2$.

Pseudo-Metric on $\mathbb{R}^2$

Pseudo-Metric on $\mathbb{R}^2$

The pseudo-metric on $\mathbb{R}^2$, is obtained by taking the set of ordered pairs of real numbers,

$$x = (\xi_1, \xi_2), \qquad y = (\eta_1, \eta_2),$$

and defining the *Pseudo-Metric* by

$$d(x, y) = |\xi_1 - \eta_1|.$$

Show that d is a pseudo-metric on $\mathbb{R}^2$.

Metric Space $\mathbb{R}^n$

Standard Euclidean Metric in $\mathbb{R}^n$

The metric space $\mathbb{R}^n$ is obtained by taking the set of ordered n-tuples of real numbers,

$$x = (\xi_1, \xi_2, \ldots, \xi_n), \qquad y = (\eta_1, \eta_2, \ldots, \eta_n),$$

and defining the *Euclidean metric* by

$$d(x, y) = \left(\sum_{i=1}^{n} |\xi_i - \eta_i|^2 \right)^{\frac{1}{2}}$$

Show that d is a metric in $\mathbb{R}^n$.

d_p Metric in $\mathbb{R}^n$

The metric space $\mathbb{R}^n$ is obtained by taking the set of ordered n-tuples of real numbers,

$$x = (\xi_1, \xi_2, \ldots, \xi_n), \qquad y = (\eta_1, \eta_2, \ldots, \eta_n),$$

and defining the *Euclidean metric* by

$$d_p(x, y) = \left(\sum_{i=1}^{n} |\xi_i - \eta_i|^p \right)^{\frac{1}{p}} \qquad (p \geq 1).$$

Show that d_p is a metric in $\mathbb{R}^n$.

d_∞ Metric in $\mathbb{R}^n$

The metric space $\mathbb{R}^n$ is obtained by taking the set of ordered n-tuples of real numbers,

$$x = (\xi_1, \xi_2, \ldots, \xi_n), \qquad y = (\eta_1, \eta_2, \ldots, \eta_n),$$

and defining the *Euclidean metric* by

$$d_\infty(x, y) = \max\{|\xi_1 - \eta_1|, |\xi_2 - \eta_2|, \ldots, |\xi_n - \eta_n|\}.$$

Show that d_∞ is a metric in $\mathbb{R}^n$.

Theorem

Prove that the function $d_\infty : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ defined by

$$d'(x, y) = \max_{1 \le i \le n} |x_i - y_i|$$

where $x = (x_1, x_2, \ldots, x_n),\ y = (y_1, y_2, \ldots, y_n) \in \mathbb{R}^n$, is a metric on $\mathbb{R}^n$. Also prove that

$$d_\infty(x, y) \le d(x, y) \le \sqrt{n}\, d_\infty(x, y), \qquad \forall\, x, y \in \mathbb{R}^n,$$

where d denotes the standard Euclidean metric on $\mathbb{R}^n$.

Metric Space $\mathbb{C}$

Standard metric on $\mathbb{C}$

Prove that the function $d : \mathbb{C} \times \mathbb{C} \to \mathbb{R}$ defined by

$$d(z_1, z_2) = |z_1 - z_2| \qquad \forall\, z_1, z_2 \in \mathbb{C},$$

forms a metric on $\mathbb{C}$.

A metric on $\mathbb{C}$

Prove that the function $d : \mathbb{C} \times \mathbb{C} \to \mathbb{R}$ defined by

$$d(z_1, z_2) = \frac{|z_1 - z_2|}{\sqrt{1 + |z_1|^2}\sqrt{1 + |z_2|^2}}, \qquad \forall\, z_1, z_2 \in \mathbb{C},$$

forms a metric on $\mathbb{C}$.

Metric Space $\mathbb{C}^n$

Metric in $\mathbb{C}^n$

Prove that the function $d : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{R}$ defined by

$$d(z, w) = \sqrt{\sum_{k=1}^{n} |z_k - w_k|^2}$$

where $z = (z_1, z_2, \ldots, z_n),\ w = (w_1, w_2, \ldots, w_n) \in \mathbb{C}^n$, forms a metric in $\mathbb{C}^n$.

Function Space $C[a, b]$

Max Metric on $C[a, b]$

Prove that the function $d : C[a, b] \times C[a, b] \to \mathbb{R}$ defined by

$$d(f, g) = \max_{x \in [a,b]} |f(x) - g(x)|$$

where $f, g \in C[a, b]$, forms a metric on $C[a, b]$.

Integral Metric on $C[a,b]$

Prove that the function $d : C[a,b] \times C[a,b] \to \mathbb{R}$ defined by

$$d(f,g) = \int_a^b |f(x) - g(x)| \, dx$$

where $f, g \in C[a,b]$, forms a metric on $C[a,b]$.

Space of Bounded Functions $B(A)$

Supremum Metric on $B(A)$

Prove that the function $d : B(A) \times B(A) \to \mathbb{R}$ defined by

$$d(f,g) = \sup_{x \in A} |f(x) - g(x)|$$

where $f, g \in B(A)$, forms a metric on $B(A)$.

Hamming Distance

Hamming Distance

Prove that the function $d : X^n \times X^n \to \mathbb{R}$ defined by

$$d(x, y) = \sum_{i=1}^{n} \delta(x_i, y_i)$$

where

$$\delta(a, b) = \begin{cases} 0, & \text{if } a = b; \\ 1, & \text{if } a \neq b, \end{cases}$$

and $x = (x_1, x_2, \ldots, x_n),\ y = (y_1, y_2, \ldots, y_n) \in X^n$, forms a metric on X^n.

Thank You!

We'd love your questions and feedback.

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References

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